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SUPERSEDING
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MILITARY SPECIFICATION

GENERATOR, RINGING, HAND G-42( )/PT

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of 20-cycle hand generator designated as Generator, Ringing, Hand G-42( )/PT.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-S-571 — Solder; Soft (Tin Tin-Lead, and Lead Silver).

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

LLL-F-291 — Fiberboard; Corrugated, Single Face, Flexible.

PPP-B-566 — Boxes, Folding Paperboard.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed, (Lock-Corner).

PPP-B-636 — Boxes, Fiber, Solid (for Domestic Use).

MILITARY

MIL-P-14 — Plastic-Materials, Molding and Plastic-Parts, Molded; Thermosetting.

MIL-P-79 — Plastic-Materials, Laminated, Thermosetting; Rods and Tubes.

MIL-P-116 — Preservation, Methods of.

MIL-V-173 — Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.

MIL-V-1137 — Varnish, Electrical-Insulating (for Electromotive Equipment).

FSC 5805

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MIL-P-3115 — Plastic - Material, Laminated, Thermosetting: Sheets, Paper-Base, Phenolic-Resin.

MIL-F-14072 — Finishes for Ground Signal Equipment.

STANDARDS

FEDERAL

FED-STD-595 — Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

MIL-STD-169 — Extreme - Temperature Cycle.

MIL-STD-170 — Moisture Resistance Test Cycle for Ground Signal Equipment.

MIL-STD-202 — Test Methods for Electronic and Electric Component Parts.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects for.

DRAWINGS

SIGNAL CORPS

SC-DL-108591 — Generator, Ringing, Hand G-42( )/PT.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions, may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following publications of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

NATIONAL BUREAU OF STANDARDS

Handbook H-28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification.
Consolidated Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Description. Generator, Ringing, Hand G-42( )/PT consists of a permanent magnet armature and coils mounted on a laminated field. A centrifugally operated switch is incorporated which operates at a crank speed of about 140 r.p.m. At a crank speed of 200 r.p.m., the generator delivers 100 volts at 20 cycles. The overall size of the generator, including the crank assembly, is $3\frac{3}{8}$ inches by $2\frac{9}{16}$ inches.

3.2 Construction. Generator, Ringing, Hand G-42( )/PT shall be constructed in accordance with the Drawing and Data List SC-DL-108591 and shall meet the requirements of this specification. In case of a conflict between the specification and the drawings, the specification shall govern.

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3.3 Preproduction samples. The contractor shall furnish preproduction samples for approval, including circuit diagram and pilot pack, as required by the invitation for bids and contract. (See 6.2.)

3.4 Parts, materials, and processes, general.

3.4.1 *Plastic material.* Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the kind of material used and the intended application.

3.4.1.1 *Thermosetting.* Thermosetting plastic material shall conform to table I and, unless hermetically sealed inside of parts, shall be treated in accordance with table II.

TABLE I. *Thermosetting plastic material*

Material	Specification
Laminated rods and tubes	*MIL-P-79
Laminated sheets, paper-base, phenolic-resin	MIL-P-3115
Molding	MIL-P-14

*Cotton-fabric-base laminates shall not be used unless high impact resistance is required for gears, cams, or similar applications or unless specified herein for a particular application.

TABLE II. *Treatment of thermosetting plastic materials.*

Material	Treatment
CMG, CMI-5, MAG, MAI-60, MDG, MFE, MME, MMI-30 per Specification MIL-P-140. PBE, GMG per Specification MIL-P-79. PBE, PBE-P per Specification MIL-P-3115.	Do not treat.
All unmachined, molded materials except those listed above.	Treat with varnish conforming to Specification MIL-V-173.
All laminated materials and all machined, molded material, except those listed above.	Treat all surfaces with insulating varnish type M, grade CB, conforming to Specification MIL-V-1137, and bake for the time and at the temperature specified by the manufacturer of the varnish.

3.4.2 *Screw threads.* Screw threads shall conform to Handbook H-28 and shall be class 2 fit. Except for holding labels or similar lightweight parts not used for structural purposes or critical applications, threads in aluminum or other soft material shall not be employed where other means of fastening are practicable and in particular shall not be used when frequent disassembly or reassembly may be necessary.

3.4.3 *Soldering.*

3.4.3.1 *Solder.* Solder used for electrical connections shall be Sn60 conforming to Specification QQ-S-571.

3.4.3.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material.

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Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.3.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystalized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.4 *Cleaning.* The equipment shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.4.5 *Finish, protective.* The equipment shall be given protective finish in accordance with Specification MIL–F–14072. The final paint film on type I surfaces shall be with semigloss enamel, color No. 27038 of Standard FED–STD–595.

3.4.6 *Marking.*

3.4.6.1 *General.* Marking shall conform to Standard MIL–STD–130.

3.4.6.2 *Serial numbers.* Serial numbers are not required.

3.5 Electrical and mechanical requirements.

3.5.1 *Electrical requirements.*

3.5.1.1 *Insulation.* The insulation resistance between terminals and cover shall not be less than 1 megohm at any time (see 4.5.1.2).

3.5.1.2 *Open circuit voltage.* The open circuit voltage at 20 cycles shall be at 100 ±5 volts r.m.s. (see 4.5.1.3).

3.5.1.3 *Power output.* The power output, at a generated frequency of 20 cycles, shall be not less than 1.75 watts (see 4.5.1.4).

3.5.1.4 *Over speed.* With crank speed increased to give 30 cycles per second the power output shall not fall below the rated value (see 4.5.1.5).

3.5.1.5 *Harmonic content.* The harmonic content at a generated frequency of 20 cycles shall be not greater than 10 percent (see 4.5.1.6).

3.5.2 *Mechanical requirements.*

3.5.2.1 *Operations.* The generator shall be capable of withstanding a minimum of 200,000 operations as specified in 4.5.2.1. After 200,000 operations the starting torque shall not have increased by more than 10 percent and the generator shall meet the requirements of 3.5.1.2, 3.5.1.5, and 3.5.2.5 (see 4.5.2.1).

3.5.2.2 *Frequency.* A generated frequency of 20 cycles per second shall be obtained at a crank speed of 200 ±5 r.p.m. (see 4.5.2.2).

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3.5.2.3 *Starting torque.* The starting torque shall not exceed 2½ inch pounds in either a clockwise or counter-clockwise direction (see 4.5.2.3).

3.5.2.4 *Switch operation.* The switch, controlled by the operating mechanism shall be in one position when the generator is idle and the other position during operation. The switch shall operate when the generator handle attains a crank speed of between 110 and 125 r.p.m. The spring tension shall be not less than 30 grams during operation, and with the unit oriented in any position (see 4.5.2.4).

3.5.2.5 *Switch click elimination.* When the generator is permitted to coast to a stop from normal operating speed (20 c.p.s.), the switch assembly shall prevent the normally closed switch contacts from making contact until the armature comes to a complete stop. The contacts shall be closed before 0.5 seconds after the armature stops rotating (see 4.5.2.5).

3.6 Service conditions. The equipment shall meet the following service conditions:

3.6.1 *Equipment nonoperating.* The equipment shall comply with the operational requirements of 3.5.1.1, 3.5.1.2, 3.5.1.5, and 3.5.2.4 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) *Temperature.* Exposure at +160° F. and at —80° F. in accordance with steps 2A and 6A of Standard MIL-STD-169 (see 4.6.1).

(b) *Corrosion resistance.* There shall be no evidence of corrosion and the markings shall remain legible when generators are tested as specified in 4.6.2.

(c) *Elevation.* Elevations up to 50,000 feet above sea level (see 4.6.3).

(d) *Immersion.* When tested in accordance with 4.6.4 the sealed units of the generator shall show no evidence of water penetration (see 4.6.4).

3.6.2 *Equipment operating.* The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Temperature.* Any ambient temperature in the range of ±125°F. to —40° F. (see 4.6.1).

(b) *Relative humidity.* Any relative humidity up to 100 percent (see 4.6.5).

3.7 Shock, drop. After subjection to the shock test of 4.6.6 the equipment shall meet the requirements of the following paragraphs:

(a) Open circuit voltage.....3.5.1.2

(b) Switch operation 3.5.2.4

(c) Torque 3.5.2.3

3.8 Preconditioning. The equipment shall be capable of meeting the inspections of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.9 Adjustment and repair. The equipment shall be so constructed that no special tools are required for adjustment and repair.

3.10 Interchangeability. Corresponding units and replaceable assemblies, subassemblies and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used (see 4.7). If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

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3.11 Workmanship. The equipment shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with the Drawing and Data List SC-DL-108591 and the applicable portions of the following paragraphs herein:

Construction 3.2
Soldering 3.4.3
Cleaning 3.4.4
Finish, protective 3.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 *Classification of inspection.* Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified in 4.3 and 4.9.)

1. Procurement inspection of items before preparation for delivery (see 4.3).

2. Procurement inspection of preparation for delivery (see 4.9 and section 5).

4.2 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III and the nondestructive group A, group B, and group C examinations and tests specified in tables IV, V, and VI respectively. The preproduction inspection will normally be performed in this order: (1) Shock, drop, (2) Immersion; other preproduction inspection may precede, follow or be interspersed between the foregoing.

TABLE III. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.2)	Requirement par.	Inspection par.
Altitute	3.6.1(c)	4.6.3
Corrosion	3.6.1(b)	4.6.2
Humidity	3.6.2(b)	4.6.5
Temperature	3.6.1(a) and 3.6.2(a)	4.6.1

4.3 Procurement inspection of items before preparation for delivery. The contractor shall perform the inspection specified by (a) through (d) below, to demonstrate compliance with the specified requirements. However, this does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the inspection specified in (a), to verify the contractor's compliance with specified requirements. (The amount of such verification inspection will depend on the extent to which the contractor's inspection procedures and records insure that the equipment on contract meets the specified procurement inspection. Also see 6.7.)

(a) Procurement inspection shall consist of group A, group B, and

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group C inspection as specified in 4.3.1 through 4.3.3.2.

(b) When Standard MIL–STD–105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly (see 3.8).

4.3.1 *Group A inspection.* This inspection (including sampling) shall conform to table IV and Standard MIL–STD–105. Normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE IV. *Group A inspection*

Inspection or test	Requirement par.	Inspection par.	AQL	
			Major	Minor
Visual and mechanical:				
Click elimination	3.5.2.5	4.5.2.5	1.0 percent for the group.	4.0 percent for the group*.
Starting torque	3.5.2.3	4.5.2.3		
Switch operation	3.5.2.4	4.5.2.4		
Electrical:				
Insulation	3.5.1.1	4.5.1.2	1.0 percent for the group.	**
Open circuit voltage	3.5.1.2	4.5.1.3		
Power output	3.5.1.3	4.5.1.4		
Overspeed	3.5.1.4	4.5.1.5		

* Failure to meet a requirement of either starting torque or switch operation is a major defect.

** All electrical defects are considered major.

4.3.2 *Group B inspection.* This inspection (including sampling) shall conform to table V and the appendix to Standard MIL–STD–105. Normal inspection shall be used at the start of the contract. Reduced inspection procedure shall be R–1. The AQL shall be 6.5 percent defective and the inspection level shall be L–7 for normal and tightened inspection and L–5 for reduced inspection. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL–STD–105 for disposition of the rejected product.

4.3.3 *Group C inspection.* This inspection shall be as listed in table VI.

4.3.3.1 *Sampling for inspection of equipment.* One each generator per 1000 units or one generator per month, whichever occurs sooner, shall be selected without regard to their quality for each group C inspection.

4.3.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the cor-

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TABLE V. *Group B inspection*

Inspection	Requirement par.	Inspection par.	Sampling plan
Harmonic content	3.5.1.5	4.5.1.6	6.5 percent defective L–7 N & T, L–5 R
Frequency output	3.5.2.2	4.5.2.2	
Immersion	3.6.1(d)	4.6.4	6.5 percent defective L–7 N & T, L–5 R
Interchangeability	3.10	4.7	6.5 percent defective L–7 N & T, L–5 R

rective action taken on the process and all units of the product which were manufactured with the same conditions, materials, processes, etc. (If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer) (see 6.4).

TABLE VI. *Group C inspection*

Inspection	Requirement par.	Inspection par.
Operations	3.5.2.1	*4.5.2.1
Shock; drop	3.7	4.6.6

*Sample units subjected to this inspection shall be furnished on contract.

4.3.4 *Reinspection of conforming group B and Group C sample units.* Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.5 *Disposition of nonconforming product.* When defective sample units or failed inspection lots are submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.4 Bounce preconditioning. The equipment shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the equipment shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Electrical and mechanical tests.

4.5.1 *Electrical tests.*

4.5.1.1 *Test set up.* Couple the generator crank to an adjustable speed driver source. Connect a 1,500 plus or minus 5 percent ohm resistor across the generator output terminals. Use a Ballantine model 300 voltmeter, or equal, to measure the output voltage, and a General Radio type 736A Wave Analyzer, or equal, to determine the harmonic content.

4.5.1.2 *Insulation test.* The insulation resistance between terminals and cover shall be tested for compliance with 3.5.1.1.

4.5.1.3 *Open circuit voltage test.* With the speed of the generator adjusted so the gen-

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erator frequency is 20 cycles per second, measure the open circuit voltage of the generator with the Ballantine model 300 voltmeter or equal. The open circuit voltage shall be 100 plus or minus 5 volts. (See 3.5.1.2.)

4.5.1.4 *Power output test.* With the speed of the generator adjusted so that the generated frequency is 20 cycles per second, the power output shall be at least 1.75 watts (see 3.5.1.3).

4.5.1.5 *Overspeed test.* Adjust the speed of the generator so the generated frequency is 30 cycles per second. The power output shall not be less than 1.75 watts (see 3.5.1.4).

4.5.1.6 *Harmonic content test.* With the speed of the generator adjusted so the frequency is 20 cycles per second, measure the harmonic output using a General Radio Type 736A Wave Analyzer, or equal. The harmonic content shall not exceed 10 percent (see 3.5.1.3).

4.5.2 *Mechanical tests.*

4.5.2.1 *Operations.* Each operation shall consist of rapidly accelerating the crank handle in a manner similar to hand cranking, turning it for five turns, and permitting it to coast to a stop. This requires that the test feature attain a crank speed of 300 r.p.m. within 100 milliseconds as shown in figure 1 (see 3.5.2.1).

4.5.2.2 *Frequency.* Adjust the speed of the drive unit so that the generated frequency is 20 cycles per second. The crank speed shall be 200 plus or minus 5 r.p.m. (see 3.5.2.2).

4.5.2.3 *Starting torque.* The starting torque shall be measured using an Apco Mossberg Co. (Attelboro, Mass.) torque tool, or equal. The starting torque shall not exceed 2½ inch-pounds (see 3.5.2.3).

4.5.2.4 *Switch operation.* Increase the speed of the driver unit until the centrifugal switch operates. The switch shall operate when the crank handle attains a speed of between 110 and 125 r.p.m. (see 3.5.2.4).

4.5.2.5 *Switch click elimination.* Permit the generator to coast to a stop from the normal operating speed (20 c.p.s). The normally closed contacts shall not make contact before the armature comes to a complete stop. The contacts shall be closed before 0.5 seconds after the armature stops rotating (see 3.5.2.5).

4.6 Service condition tests.

4.6.1 *Temperature test for equipment.* The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. The equipment shall meet the requirements of 3.5.1.2 and 3.5.2.5 at Step 4, Step 8, and at Step 10 of the temperature cycle (see 3.6.1(a) and 3.6.2(a)).

4.6.2 *Corrosion test.* The generator, in operating condition, shall be subjected to a 20 percent salt spray in accordance with Method 101 of Standard MIL-STD-202, test condition A. After removing any salt accumulations and drying out the generator, examination shall show no harmful corrosive action or other damage to any part due to the salt spray, and the equipment shall meet the requirements of 3.5.1.1, 3.5.1.2, and 3.5.2.3 (see 3.6.1(b)).

4.6.3 *Elevation test.* The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure and humidity. Performance shall be measured. The pressure shall then be reduced to 20 inches of mercury and stabilized for two hours. The performance of the equipment shall again be measured. The pressure shall then be lowered to 9 inches of mercury and the chamber maintained at this pressure for another 2-hour period. The pressure shall then be increased to 29.9 inches of mercury

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STARTING CHARACTERISTIC OF GENERATOR G-42( )/PT FOR
HANDSPUN OPERATION WITH 1500 OHM LOAD

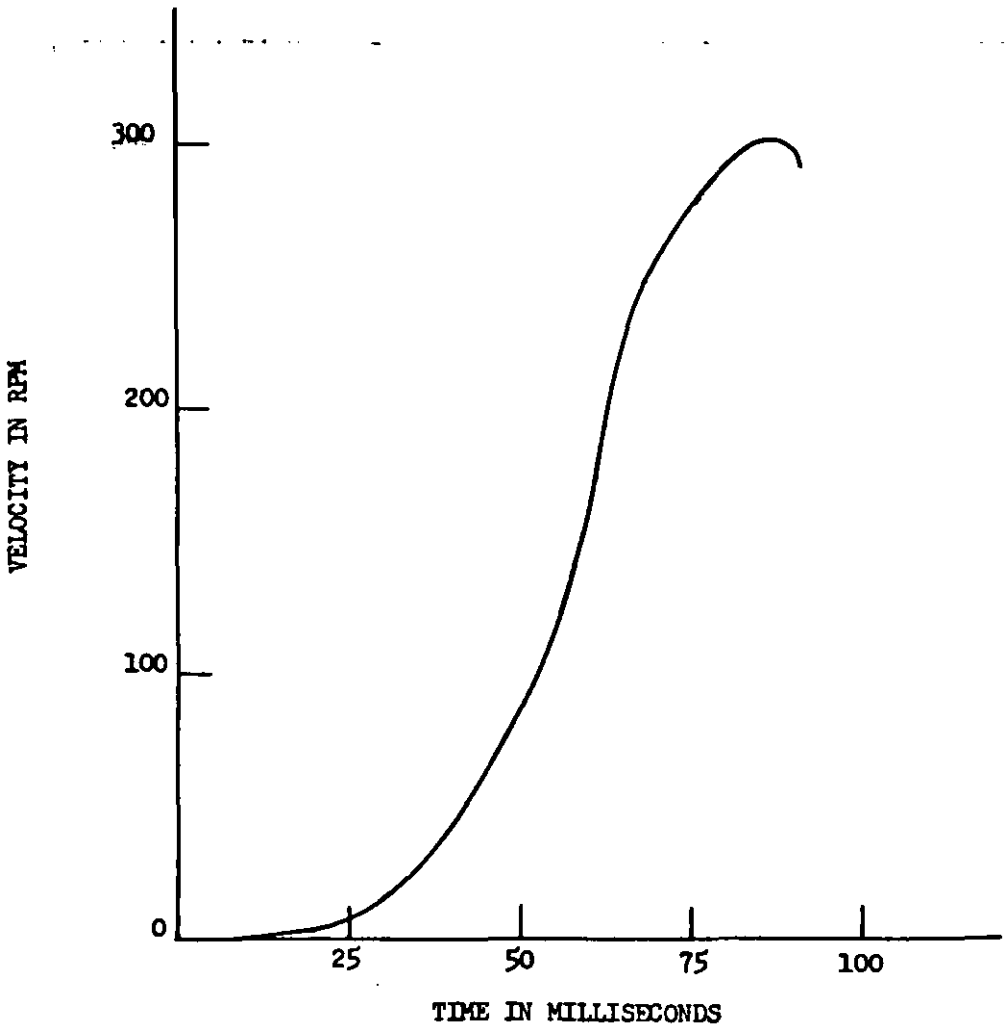


FIGURE 1. *Generator, ringing, hand G-42( )/PT.*

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and the performance measured again. There shall be no degradation of performance in the second and third measurements from those established in the first performance measurement (see 3.6.1(c)).

4.6.4 *Immersion test.* The equipment shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40° F., or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2° F. of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage (see 3.6.1(d)).

4.6.5 *Moisture-resistance test for equipment.* The equipment shall be tested as follows:

4.6.5.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Complete measurements as rapidly as possible.

4.6.5.2 *Test method.*

(a) Dry at 130° ±5° F. for 24 hours.

(b) Condition at +77° ±4° F., and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 3.5.1.1 and 3.5.1.2.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurement specified in 4.6.5.3 shall be made during the periods shown on the standard and shall comply with 4.6.5.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5° F. and 40 to 50 percent relative humidity. No repair or replacement of parts shall be made. The equipment shall meet full specified performance for those measurements specified in 4.5.1.2, 4.5.1.3, 4.5.1.4, and 4.5.1.5.

4.6.5.3 *Performance.* Measurements made during test, and specified performance during cycling, shall be as follows:

Paragraph	*Measurement*	*Performance during cycling*
4.5.1.2	Insulation	Full specification performance

During the test, the measurement of 4.5.1.2 shall be made. The equipment shall meet full specification performance.

4.6.5.4 *Failure.* If the equipment fails to meet the performance specified in 4.6.5.3 during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning, as specified in 4.5.1, it does not pass the test (see 3.6.2(b)).

4.6.6 *Shock test; drop.* The equipment, mounted in a unit with which it is commonly used (such as Telephone TA-312( )/PT or the like), (see 6.2) shall be dropped a distance of 48 inches from a quick-release hook or pendulum tester as made by the L.A.B. Corporation, Skaneateles, N. Y. or equal. The floor or barrier receiving the impact shall be 2 inch fir backed by concrete or a

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rigid steel frame. The unit shall be dropped on each face, edge, and corner for a total of 26 drops. The equipment shall be operable, with no serious degradation in performance electrically or mechanically at the completion of the test. Minor mechanical damage consistent with the preceding is permissible (see 3.7).

4.7 Inspection for interchangeability. The dimensions of the parts shall be gaged or measured to determine compliance with the drawings on Data and Drawing List SC-DL-108591. When a dimension is not within the limits specified on the drawing it shall be considered a major defect (see 3.10).

4.8 Visual and mechanical inspection. A thorough inspection shall be made on each completed equipment for mechanical fit, loose nuts and screws, finish, poor workmanship and other defects. Fastening devices shall be checked for proper mechanical operation. Finishes, connections and the like shall be checked for satisfactory workmanship and compliance with Standard MIL-STD-252.

4.9 Acceptance inspection of preparation for delivery.

4.9.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to tables VII and VIII.

TABLE VII. *Classification of defect for preservation, packaging and marking*

Major defects

1. Quantity per unit package not in accordance with the contract.
2. Incorrect packaging method applied.
3. Use of improper or defective material.
4. Cushioning inadequate or not applied when required for the physical or mechanical protection of the item.
5. Stock number omi .ed, incorrect or illegible.
6. Nomenclature incorrect or missing.
7. Quantity of items in package not indicated.

Minor defects

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Any item of marking information other than 5 through 7 listed under major defects missing or illegible.

TABLE VIII. *Classification of defects for packing and marking*

Major defects

1. Box closure not in accordance with specification.
2. Type, grade, class, and style of the shipping container do not meet the minimum requirements.
3. Strapping, when required, is omitted or inadequate, or incorrectly applied.
4. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
5. Gross weight in excess of contract limitation.
6. Use of improper or defective material.
7. Shipping documents or packing list omitted (when required).
8. Stock number (when required) omitted, incorrect, or illegible.
9. Nomenclature (when required) omitted or illegible.
10. Quantity incorrect or missing.
11. Destination marking incorrect or illegible.
12. Service designation (color marking) missing.
13. Special marking and labeling requirements cited in the contract, such as MAP labels, shipment digit markings, etc., not applied.
14. Overseas code marking illegible, incomplete or missing.

Minor defects

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.

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4. Any item of required marking information, other than 8 through 14 listed under major defects, missing or illegible.

4.9.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed in table VII and table VIII and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent.

4.9.3 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.9.4 *Disposition of nonconforming product.* Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL–STD–105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Generator, Ringing, Hand G–42( )/PT shall be preserved, packaged, and tested in accordance with the procedure specified for the designated methods as prescribed in Specification MIL–P–116 and described herein.

5.1.1.1 *Generator, Ringing, Hand G–42 ( )/PT.* Each generator shall be packaged Method III as follows: Cushion each generator within a wrap of flexible, single-faced, corrugated fiberboard, Type II, conforming to Specification LLL–F–291. Secure cushioning with Class 2, gummed paper tape, conforming to Specification UU–T–111. Place each cushioned generator within a close-fitting, folding paperboard box, Style II, Type A, class a, conforming to Specification PPP–B–566. Box closure shall be in accordance with the requirements of the box specification.

5.1.1.2 *Intermediate container.* A quantity of generators, packaged as specified in 5.1.1.1 shall be placed within a regular slotted style, Type I, Grade 5, fiberboard box conforming to Specification PPP–B–636. Fabricate the box to fit the contents snugly. The gross weight of any intermediate container shall not exceed 20 pounds. Seal the box as specified in the appendix of the box specification.

5.1.2 *Level C.* Generator, Ringing, Hand G–42( )/PT shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* A quantity of intermediate containers containing Generator, Ringing, Hand G–42( )/PT shall be packed within a class 2, nailed wood box, conforming to Specification PPP–B–621 or an overseas type, wood cleated-plywood box conforming to Specification PPP–B–601. Fabricate the box to fit the contents snugly. The gross weight of any shipping container shall not exceed 200 pounds. Box closure shall be in accordance with the box specification.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat steel strapping,

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in conformance with the requirements of the appendix of the container specification only for direct shipment to ports.

5.2.2 *Level B.* Generator, Ringing, Hand G-42( )/PT shall be packed as specified in 5.2.1 except that the nailed wood box shall conform to Class 1, Style 4, of Specification PPP-B-621 and the wood cleated-plywood box shall conform to domestic type of Specification PPP-B-601. No metal strapping is required.

5.2.3 *Level C.* Generator, Ringing, Hand G-42( )/PT shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed at the lowest transportation rate.

5.3 **Marking.** Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Generator, Ringing, Hand G-42( )/PT provides a 20 cycle ringing signal for field telephone sets, switchboards, and telephone circuit control boxes. It may be used in applications which require a panel mounted generator.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of the specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(d) Preproduction samples required. Two of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (see 4.2).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required.

(h) Quantity of tools and running spares required.

(i) Detailed instructions for preparation for delivery.

(*Note.* These may be obtained from Signal Corps Packaging Activity, U. S. Army Signal Supply Agency, 225 Scuth Eighteenth Street, Philadelphia 3, Pa.)

(j) Government furnished equipment (see 4.6.6).

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: G-42W/PT. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.3.5).

6.5 Definition of level B. Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.6 Inspection terms. The inspection terms used herein conform to Standard MIL-STD-109, Inspection Terms and Definitions.

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6.7 **Verification inspection.** The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) *Type A.* The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL–STD–105.

(b) *Type B.* That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

(c) *Type C.* A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based on a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Yards and Docks
Air Force

Preparing activity:
Army—Signal Corps

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